



452 THE PASSENGER CAPACITY OF THE BOEING 747-100

The first Boeing 747-100—*Clipper Victor*—went into commercial service on January 22, on a PanAm flight from New York to London.

THE AGE OF WIDE-BODY PASSENGER JET TRAVEL began when the **first Boeing 747 jumbo jet made its maiden commercial voyage** in January of 1970. The idea of the airliner was conceived in the mid-1960s, partly to ease congestion at busy airports. By 2013, more than 1,500 Boeing 747s had been built.

In February, **Japan** became the fourth country—after the USSR, the US, and France—to **send a rocket into space**, when

its national space development agency (NASDA) launched the experimental satellite called Ōsumi. **China** became the fifth country to do so, in April, with the successful launch of its experimental satellite, **Dong Fang Hong 1**.

After the successes of its first three manned missions to the Moon, **NASA's Apollo program** suffered a setback in April 1970, **during the Apollo 13 mission**. Around 55 hours after

lift off—more than 200,000 miles (320,000 km) from Earth—one of the spacecraft's two oxygen cylinders exploded. Oxygen was crucial not only to make the air in the spacecraft breathable, but also to generate electrical power and to make drinking water in the fuel cells. The mission was aborted, and the drama of the journey back to Earth was played out on radio and television. The spacecraft splashed down in the South Pacific Ocean, five days after the incident.

Later this year, NASA successfully launched Uhuru, the **first dedicated orbiting X-ray observatory**. X-ray astronomy is only possible at high altitude—ideally, in orbit—because the atmosphere absorbs most of the X-ray radiation from space.

The first Earth Day—an annual celebration of the world's natural environment and a call to action for environmentalists to protect it—was observed this year. It was first proposed in 1969 by American peace activist John McConnell (1915–2012), as an event that would take place on the spring equinox (Northern Hemisphere) each year—around March 21. The **first Earth Day**

Rescue mission

Three Apollo 13 astronauts are lifted aboard a helicopter in a rescue net, after their Lunar Module returned them safely to Earth on April 17.

LYNN MARGULIS (1938–2011)

American biologist Lynn Margulis is best known for her theory of complex cell evolution, which she first published in 1966 when working at Boston University. *Origin of Eukaryotic Cells* (1970), expanded on her endosymbiotic theory, but earned criticism within the scientific community. It took 30 years before sufficient evidence led to the theory's acceptance.



took place on April 22, and is seen as a major event in the history of environmentalism. It was celebrated with events across the US, but since 1990, Earth Day events have been held worldwide.

In May, American biologist **Lynn Margulis** (see panel, above) published a book expanding on her **endosymbiotic theory of the origin of eukaryotic cells** (cells with complex structures contained within a membrane), which she first proposed in 1966. Eukaryotic cells contain structures called

organelles, with specific functions (see pp.194–95); for example, plant cells contain organelles called chloroplasts, in which photosynthesis (see 1787–88) takes place. Margulis' idea was that organelles were once simple cells in their own right, and that eukaryotic cells evolved as a symbiosis of these subunits.

It was a year of important breakthroughs in genetics too. Genetic information is carried in two similar compounds inside cells: DNA and RNA (see pp.284–85). DNA stores genetic information, while RNA transfers this information and is involved in building protein molecules. Until 1970, biologists believed that information could flow in only one direction: from DNA to RNA. In June, American

“**HOUSTON, WE’VE HAD A PROBLEM.**”

John L. “Jack” Swigert, US astronaut, Apollo 13 mission, 1970



January 22
Jumbo jet enters commercial service

February 11
Japan launches first satellite, Ōsumi

April 24 China launches first satellite, Dong Fang Hong 1

April 17 NASA's Apollo 13 returns to Earth after the mission is aborted

April 22 First Earth Day is celebrated

June 19 Two cosmonauts set endurance record for human space flight

June 27 Howard Temin and American biologist David Baltimore independently discover reverse transcriptase